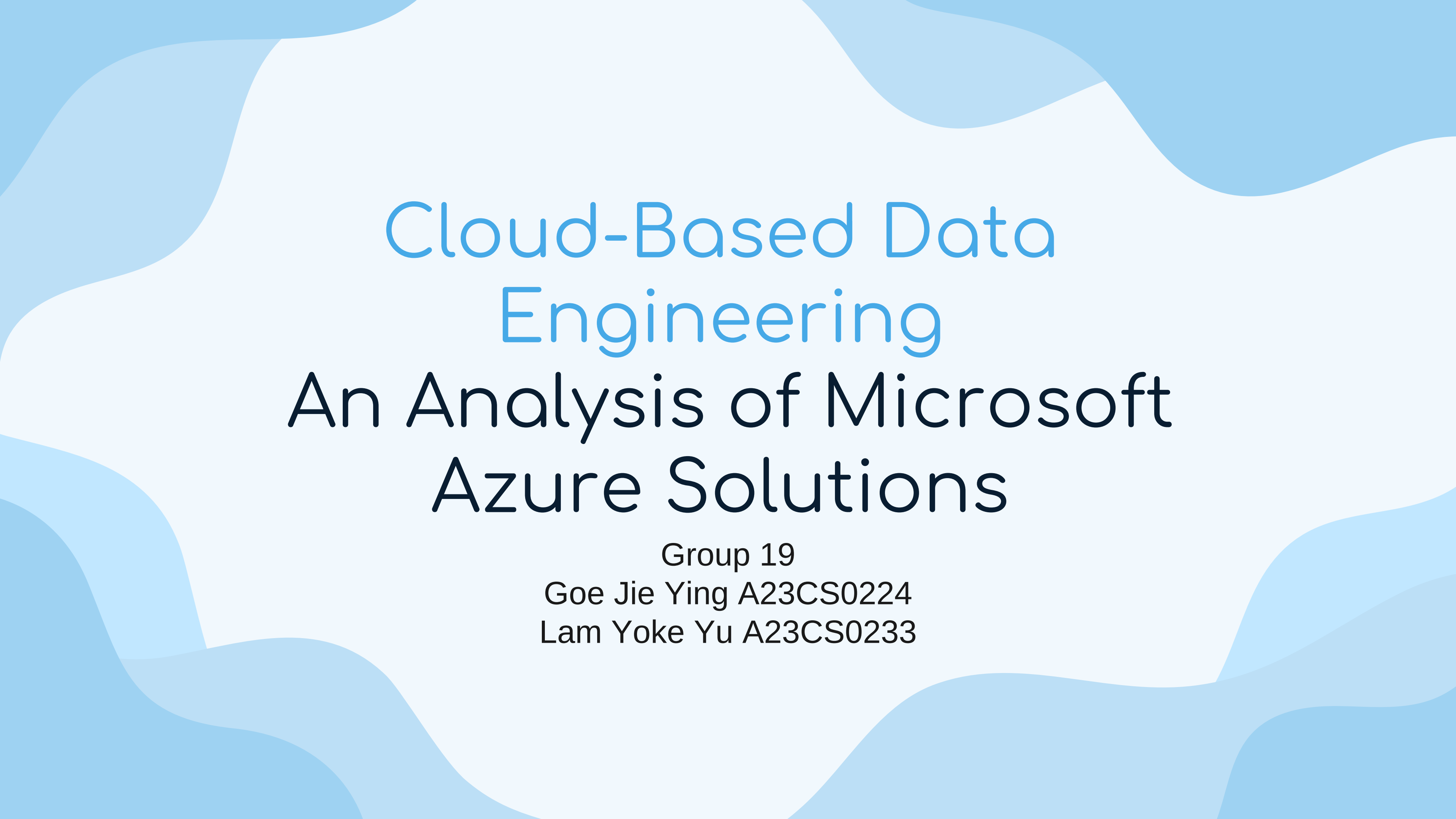




Cloud-Based Data Engineering

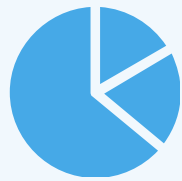
An Analysis of Microsoft Azure Solutions

Group 19

Goe Jie Ying A23CS0224

Lam Yoke Yu A23CS0233

Background



Data
Engineering

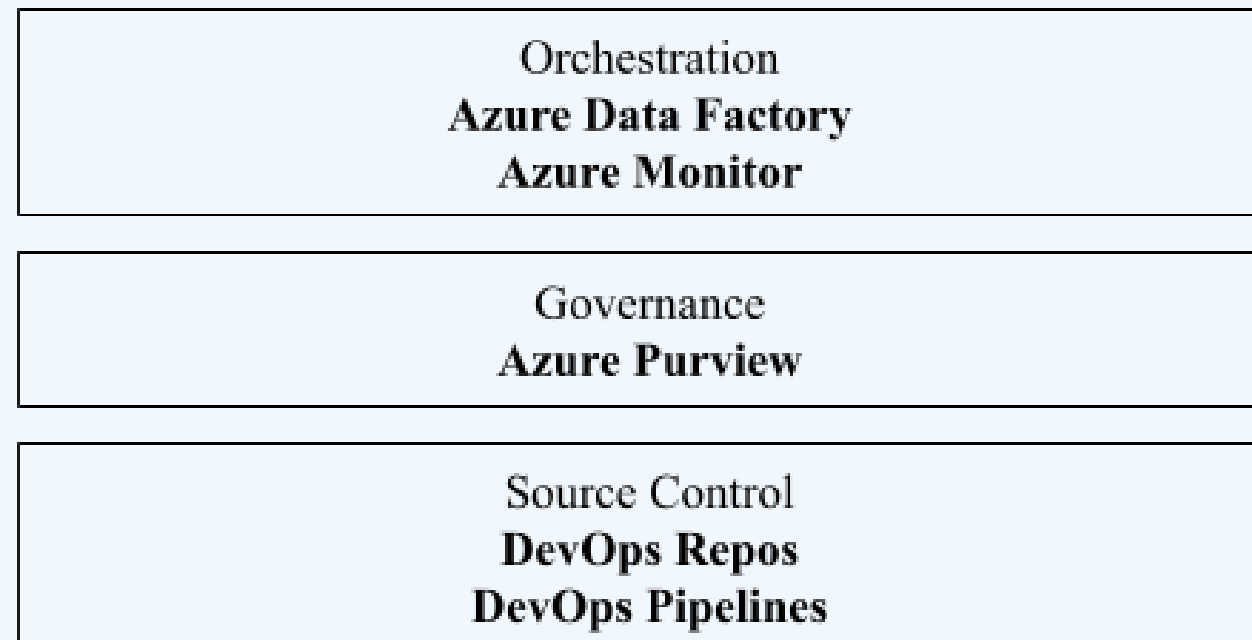
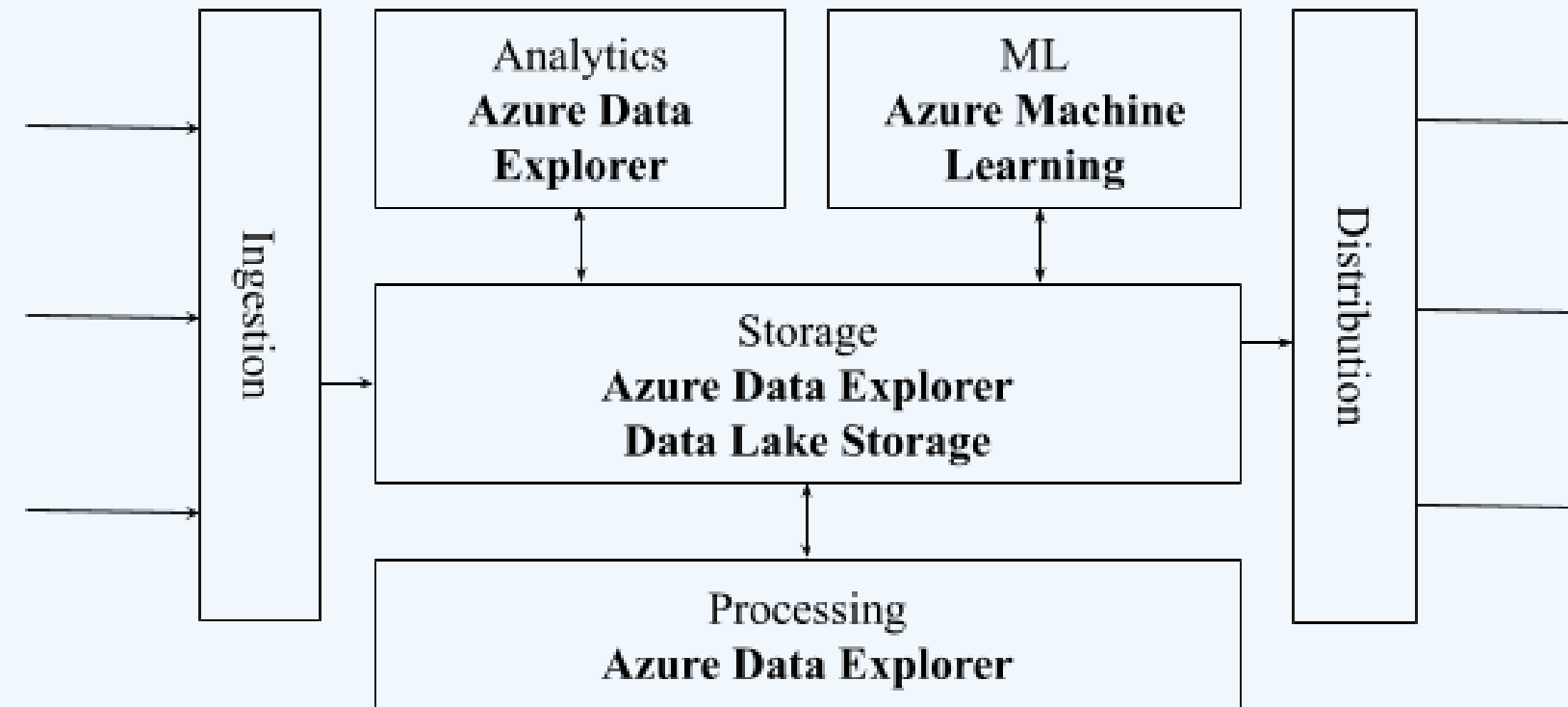


Cloud
Computing

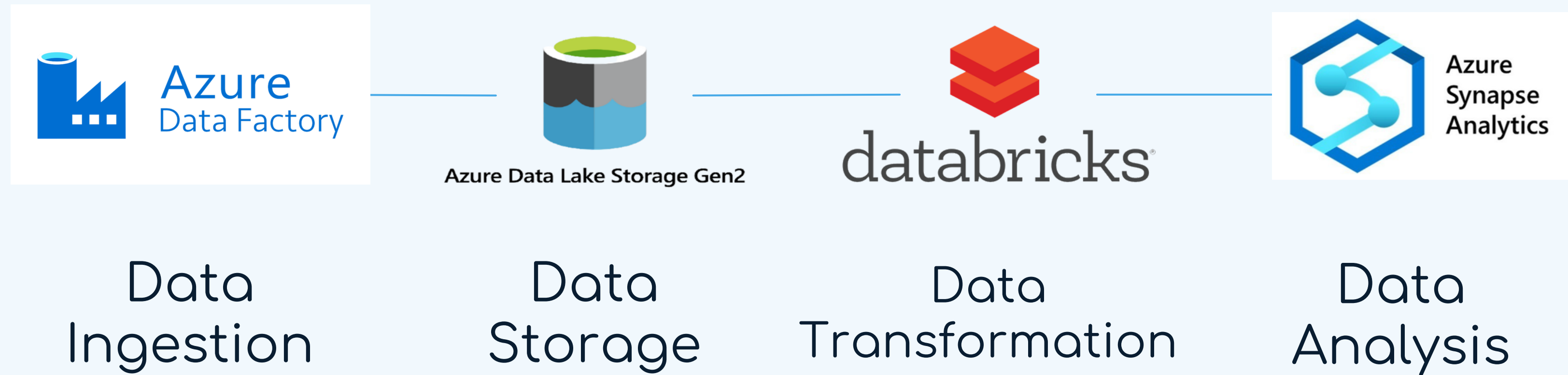


Cloud-based Data
Engineering

Microsoft Azure



Tools



Strengths



scalable and
flexible



pay-as-you-go



modern data
protection

Comparison with Other Platforms

Feature	Azure	AWS	GCP
Performance and Scalability	Mid-range, strong integration	High performance with strong reliability	Mixed, excels in specific areas
Cost Efficiency	Budget-friendly	Premium pricing	Competitive, discounts
Integration and Ecosystem	Best for Microsoft ecosystems	Versatile across industries	Scalable for data-intensive apps
Use Cases and Strengths	Enterprise integration, cost-saving	Compute-intensive, scalability	Big data, ML, latency-sensitive
Data Processing	Databricks, Synapse	EMR, Glue	Dataflow, Dataproc
Data Warehousing	Synapse	Redshift	BigQuery

Real World Application



- Azure App Service
- Azure AI
- Azure Kubernetes Service
- Azure Data Lake
- Azure Blob Storage
- Azure Synapse Analytics
- Azure Data Explorer
- Azure Databricks

Recommendation

Conclusion

Thanks

CREDITS: This presentation template was created by **Slidesgo**, including icons by **Flaticon** and infographics & images by **Freepik**